PROJECT: STREAMLINING TICKET ASSIGNMENT FOR EFFICIENT SUPPORT OPERATIONS

TEAM MEMBERS

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PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

Streamlining Ticket Assignment for Efficient Support Operations is a system developed to improve the efficiency of support services by automating the ticket assignment process. In many organizations, support requests are handled manually, where a support manager reviews each ticket and assigns it to a technician. This manual process often causes delays, incorrect assignments, and uneven workload distribution among support staff.

The proposed system automates the process using the ServiceNow platform. When a user submits a support request, the system automatically analyzes the ticket details such as category, priority, and department. Based on predefined rules, the system assigns the ticket to the appropriate support technician or support team.

This automation ensures that tickets are handled quickly and efficiently without the need for manual intervention. The system also stores all ticket information in a database, allowing administrators to track ticket progress and manage support operations effectively.

The system helps organizations improve response time, reduce manual work, and increase overall productivity. By implementing this automated ticket assignment system, support teams can focus more on resolving issues rather than spending time assigning tasks manually.

1.2 Purpose

The main purpose of this project is to simplify and improve the ticket management process in organizations. Manual ticket assignment requires time and effort, especially when the number of support requests increases. This often results in delays in resolving customer issues.

The proposed system aims to automate the process of assigning support tickets using ServiceNow workflows. When a ticket is created, the system automatically assigns it to the correct support group or technician. This ensures faster response times and better task distribution among team members.

2. IDEATION PHASE

2.1 Brainstorming – Idea Generation – Prioritization

Several ideas were discussed during the brainstorming phase. Some of the ideas included developing a digital ticket management system, creating an automated ticket routing mechanism, implementing priority-based ticket assignment, and generating automatic notifications for technicians.

Each idea was analyzed based on feasibility, effectiveness, and ease of implementation. After evaluating these ideas, the concept of automating ticket assignment using the ServiceNow platform was selected as the most suitable solution.

The automated ticket assignment system helps reduce delays in ticket handling and ensures that support requests are assigned to the correct technicians quickly. This solution improves the overall efficiency of support operations and reduces manual workload.

2.2 Define Problem Statement

In many organizations, support tickets are assigned manually by administrators or support managers. This manual process can lead to several issues such as delayed response times, incorrect ticket assignments, and uneven workload distribution among technicians.

When the number of support requests increases, it becomes difficult for administrators to manage and assign tickets efficiently. This can result in longer waiting times for users and reduced customer satisfaction.

Therefore, there is a need for an automated system that can streamline the ticket assignment process. The proposed project aims to develop a system using the ServiceNow platform that automatically assigns tickets based on predefined rules such as category, priority, or department.

2.3 Empathy Map Canvas

The empathy map canvas helps understand the needs and challenges faced by users interacting with the system. The primary users of this system are support staff, administrators, and customers who submit support requests.

Users say that manual ticket assignment takes too much time and slows down the support process. They think that an automated system would make their work easier and improve efficiency. Currently, users manually review each support request and assign it to technicians, which requires constant monitoring.

Users often feel frustrated when handling a large number of tickets because manual assignment increases workload and delays problem resolution. By understanding these challenges, the proposed system aims to automate ticket assignment and simplify support operations.

3. REQUIREMENT ANALYSIS

3.1 Data Flow Diagrams and User Stories

A Data Flow Diagram (DFD) is used to represent how data moves through the system. It shows how information is collected, processed, stored, and delivered.

In the ticket assignment system, the process begins when a user submits a support ticket through a ServiceNow form. The system processes the ticket details and determines the appropriate support team or technician based on predefined rules.

Once processed, the ticket information is stored in the ServiceNow database table. The system then sends a notification to the assigned technician, informing them about the new support request.

User stories help describe how users interact with the system.

Examples of user stories include:

- As a user, I want to submit a support ticket so that my issue can be resolved.
- As a support technician, I want tickets to be assigned automatically so that I can start resolving issues immediately.
- As an administrator, I want to monitor ticket assignments to ensure efficient support operations.

3.2 Solution Requirements

Functional Requirements

- The system should allow users to submit support tickets.
- The system should automatically assign tickets to technicians.
- The system should store ticket information in a database.
- The system should allow administrators to monitor ticket status.

Non-Functional Requirements

- The system should be easy to use.
- The system should respond quickly to user requests.
- The system should provide secure storage for ticket data.
- The system should operate reliably without errors.

3.3 Technology Stack

The project is developed using the ServiceNow platform. ServiceNow provides a cloud-based environment for building applications and automating workflows.

ServiceNow forms are used to collect ticket information from users. ServiceNow tables store ticket details such as issue description, priority level, and assigned technician. Flow Designer is used to automate ticket assignment based on predefined rules.

Notifications are also used to inform technicians when a ticket is assigned to them. These technologies work together to create an efficient ticket management system.

4. PROJECT DESIGN PHASE

4.1 Problem – Solution Fit

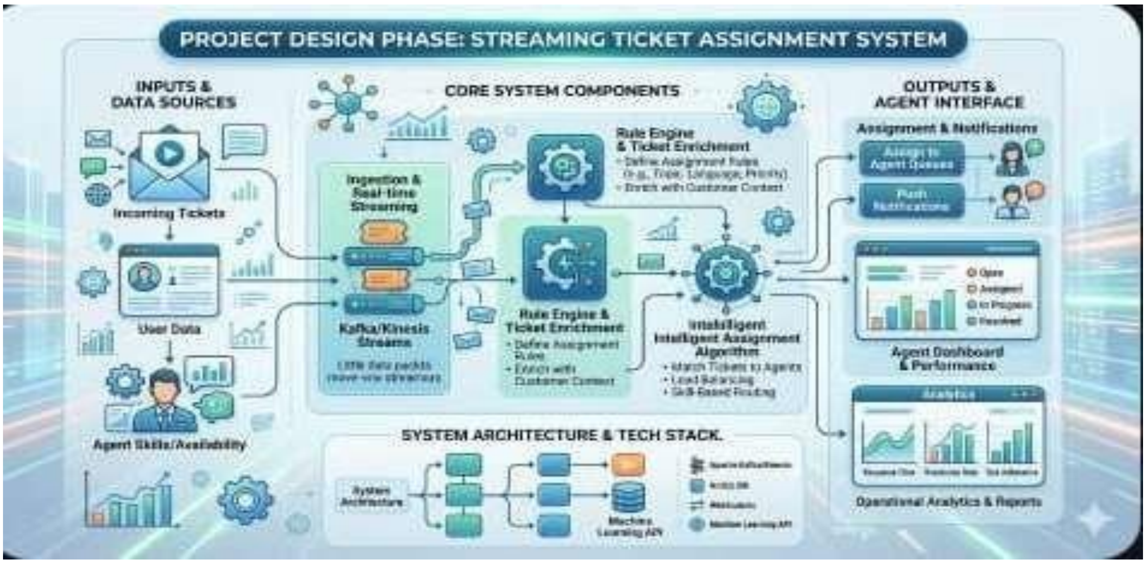
Many organizations face difficulties in managing support tickets efficiently. Manual ticket assignment often causes delays and sometimes results in incorrect assignment of tasks. As the number of support requests increases, the process becomes more complicated and timeconsuming.

The proposed system solves this problem by automating ticket assignment using the ServiceNow platform. The system allows users to submit support tickets through an online form. Once the ticket is created, the system automatically processes the information and assigns the ticket to the appropriate technician or support group.

The system architecture consists of several components. The user interacts with the system through a ServiceNow form. The submitted ticket is stored in a ServiceNow database table. Automation workflows then process the ticket and determine the correct technician for handling the issue.

The assigned technician receives a notification about the ticket and can begin working on resolving the issue. Administrators can monitor the progress of tickets and ensure that support operations run smoothly.

This structured design ensures that support tickets are handled efficiently and that users receive faster responses to their issues.


PROJECT DESIGN PHASE: STREAMING TICKET ASSIGNMENT SYSTEM
INPUTS & DATA SOURCES
Incoming Tickets
User Data
Agent Skills/Availability
CORE SYSTEM COMPONENTS
Ingestion & Real-time Streaming
Kafka/Kinesis Streams
Rule Engine & Ticket Enrichment
Rule Engine & Ticket Enrichment
Intelligent Assignment Algorithm
SYSTEM ARCHITECTURE & TECH STACK.
System Architecture
Machine Learning API
OUTPUTS & AGENT INTERFACE
Assignment & Notifications
Push Notifications
Agent Dashboard & Performance
Analytics
Operational Analytics & Reports


STREAMLINING TICKET ASSIGNMENT
FOR EFFICIENT SUPPORT OPERATIONS
NEW
IN PROGRESS
OPEN
?

5. PROJECT PLANNING PHASE

5.1 Project Planning

Project planning is an important stage in system development. It involves organizing tasks, identifying resources, and defining the steps required to complete the project successfully.

In this project, the planning process began with identifying the challenges associated with manual ticket assignment. After understanding the problem, the project objectives were defined, which included developing an automated ticket assignment system that improves support operations.

The ServiceNow platform was selected as the development platform because it provides powerful tools for workflow automation and application development. The next step involved designing the system components, including ticket submission forms, database tables, and automation workflows.

A timeline was created to guide the development process. The project was divided into different phases such as requirement analysis, system design, implementation, and testing.

Proper planning ensured that each stage of the project was completed systematically. This helped in developing an efficient system that meets the requirements of support teams and improves the ticket management process.

Planning Activities:

- Define project scope
- Prepare development timeline
- Allocate project tasks
- Identify risks
- Select development tools

6. PROJECT DEVELOPMENT PHASE

6.1 Performance Testing

During the development phase, the ticket assignment system was implemented using the ServiceNow platform. Forms were created to allow users to submit support tickets easily. These forms collect important information such as the issue description, priority level, and category.

ServiceNow tables were used to store the ticket information in a structured format. Automation workflows were then created using Flow Designer to process the tickets and assign them to the appropriate technicians.

Testing was performed to ensure that the system functions correctly. Different test cases were used to verify that tickets are stored properly and assigned automatically.

Performance testing was also conducted to evaluate the system's response time and reliability. The results showed that the system successfully assigns tickets and stores the information correctly.

The development phase demonstrated that automation can significantly improve the efficiency of ticket management systems.

7. ADVANTAGES AND DISADVANTAGES

Advantages

- Reduces manual work in ticket assignment
- Improves response time for support requests
- Ensures proper distribution of workload
- Improves efficiency of support operations
- Helps administrators track ticket progress
- Enhances customer satisfaction

Disadvantages

- Requires internet access to use the ServiceNow platform
- Initial system setup may require technical knowledge
- The system depends on the availability of the ServiceNow platform
- Organizations may need training for employees to use the system effectively

8. PROJECT DOCUMENTATION

8.1 Final Report

Project documentation provides a detailed description of the system and its development process. It includes information about the problem identified, the proposed solution, system design, development stages, and testing results.

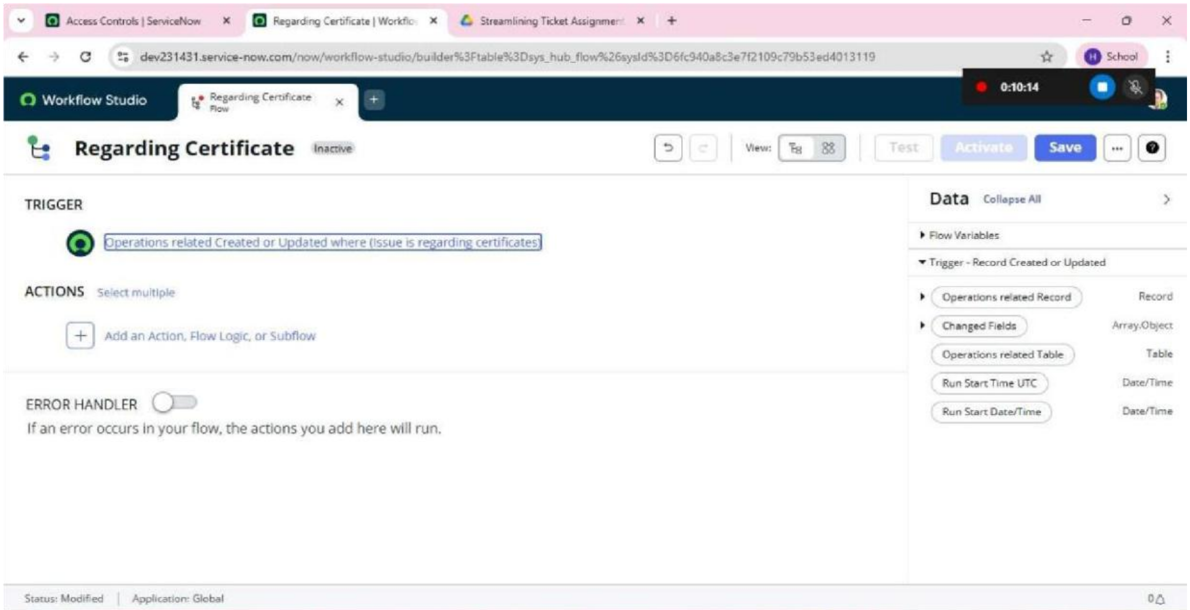
The project Streamlining Ticket Assignment for Efficient Support Operations was developed to improve the efficiency of support services by automating the ticket assignment process.

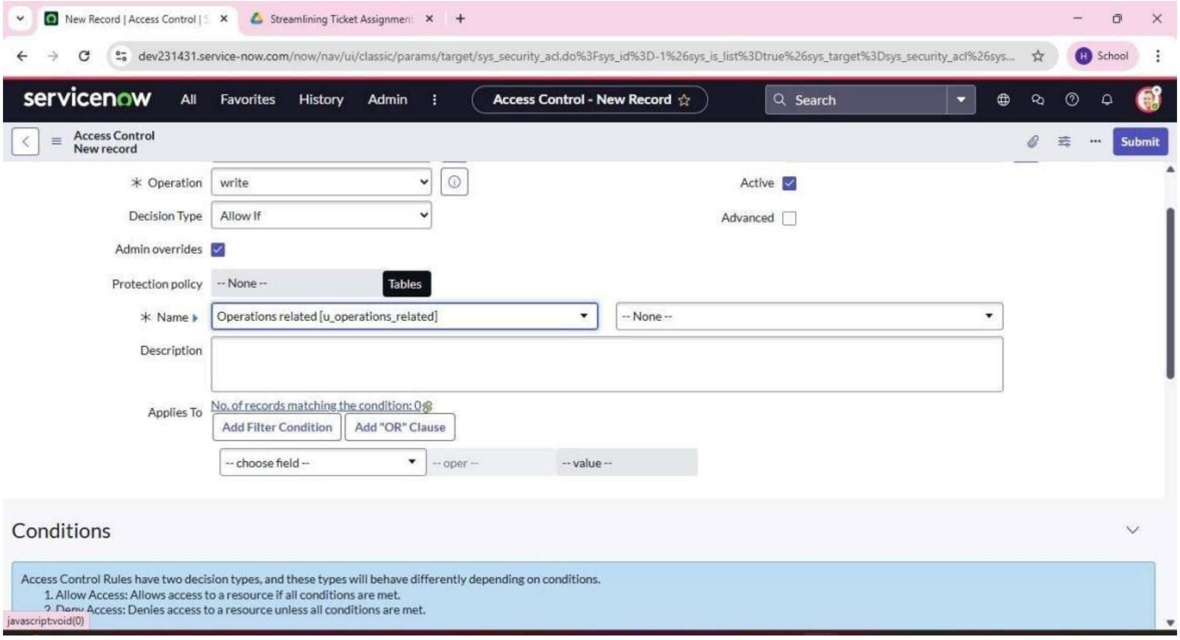
The system allows users to submit support requests through a ServiceNow form. The submitted tickets are stored in a database table and processed through automation workflows. These workflows automatically assign tickets to the appropriate support teams.

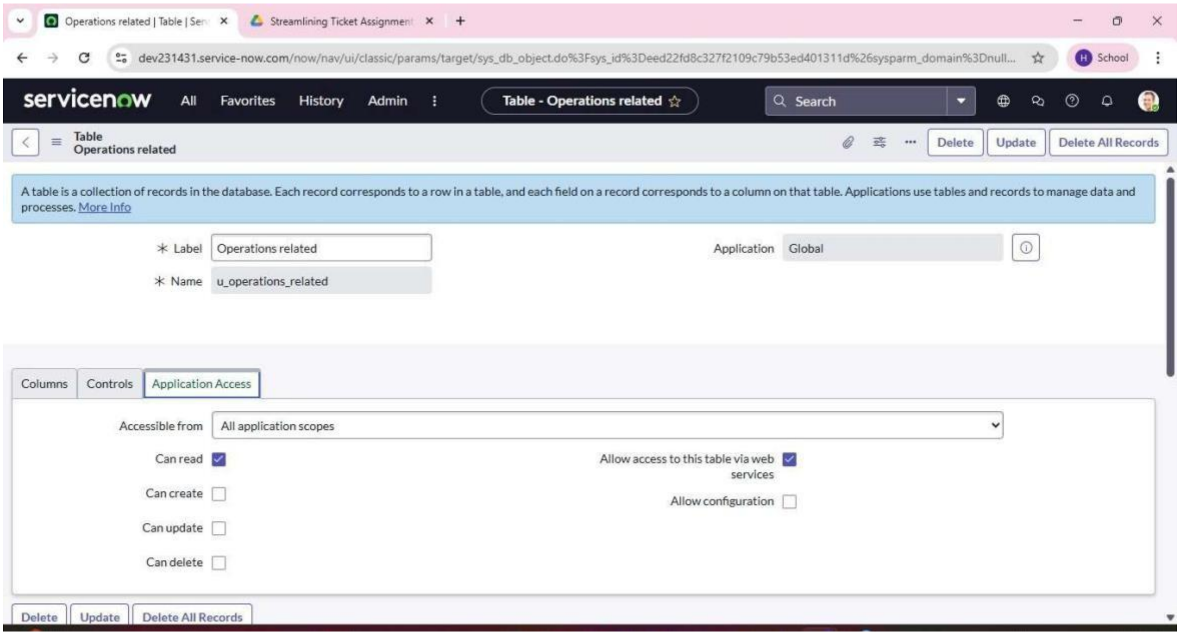
The documentation also includes diagrams such as use case diagrams, data flow diagrams, and system architecture diagrams. These diagrams help explain how the system functions and how different components interact with each other.

Screenshots of the ServiceNow interface are also included to demonstrate the working of the system. Proper documentation ensures that the project can be easily understood and maintained in the future.

8.2 Screen layout




New Record | Access Control
Streamlining Ticket Assignment
dev231431.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sys...
servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Operation
write
Active
Decision Type
Allow If
Advanced
Admin overrides
Protection policy
-- None --
Tables
Name
Operations related [u_operations_related]
-- None --
Description
Applies To
No. of records matching the condition: 0
Add Filter Condition
Add "OR" Clause
-- choose field --
-- oper --
-- value --
Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met.
2. Deny Access: Denies access to a resource unless all conditions are met.
javascript:void(0)


Operations related | Table | Ser
Streamlining Ticket Assignment
dev231431.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3Deed22fd8c327f2109c79b53ed401311d%26sysparm_domain%3Dnull...
servicenow
All
Favorites
History
Admin
Table - Operations related
Search
Table
Operations related
Delete
Update
Delete All Records
A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info
Label
Operations related
Application
Global
Name
u_operations_related
Columns
Controls
Application Access
Accessible from
All application scopes
Can read
Allow access to this table via web services
Can create
Allow configuration
Can update
Can delete
Delete
Update
Delete All Records

9. CONCLUSION

The Streamlining Ticket Assignment for Efficient Support Operations system demonstrates how automation can improve the efficiency of support services in organizations.

By using the ServiceNow platform, the system automates the process of assigning support tickets to the appropriate technicians. This reduces manual work and ensures faster response times for resolving issues.

The system also improves the management of support operations by providing administrators with tools to monitor ticket status and technician performance. As a result, organizations can handle support requests more efficiently and provide better services to their customers.

Overall, the project highlights the importance of automation in modern organizations. Implementing automated ticket assignment systems can significantly improve productivity, reduce delays, and enhance customer satisfaction